Problem Set 9
Due: Apr. 26, 2021

1. Section 4.1 Problem 13.

   Consider a PDF that is positive only within an interval $[a, b]$ and is symmetric around the mean $(a+b)/2$. Let X and Y be independent random variables that both have this PDF. Suppose that you have calculated the PDF of $X + Y$. How can you easily obtain the PDF of $X - Y$?

   Solution: We have $X - Y = X + Z - (a+b)$, where $Z = a+b-Y$ is distributed identically with X and Y. Thus, the PDF of $X + Z$ is the same as the PDF of $X + Y$, and the PDF of $X - Y$ is obtained by shifting the PDF of $X + Y$ to the left by $a + b$.

2. Section 4.2 Problem 18.

   Consider four random variables, W, X, Y, Z, with

   $$\mathbf{E}[W] = \mathbf{E}[X] = \mathbf{E}[Y] = \mathbf{E}[Z] = 0$$

   $$\text{var}(W) = \text{var}(X) = \text{var}(Y) = \text{var}(Z) = 1$$

   and assume that W, X, Y, Z are pairwise uncorrelated. Find the correlation coefficients $\rho(R, S)$ and $\rho(R, T)$, where $R = W + X$, $S = X + Y$, and $T = Y + Z$.

   Solution: We have

   $$\text{cov}(R, S) = \mathbb{E}[RS] - \mathbb{E}[R]\mathbb{E}[S] = \mathbb{E}[WX + WY + X^2 + XY] = \mathbb{E}[X^2] = 1,$$

   and

   $$\text{var}(R) = \text{var}(S) = 2,$$

   so

   $$\rho(R, S) = \frac{\text{cov}(R, S)}{\sqrt{\text{var}(R)\text{var}(S)}} = \frac{1}{2}.$$

   We also have

   $$\text{cov}(R, T) = \mathbb{E}[RT] - \mathbb{E}[R]\mathbb{E}[T] = \mathbb{E}[WY + WZ + XY + XZ] = 0,$$

   so that

   $$\rho(R, T) = 0.$$

3. Let $\{X_r : r \geq 1\}$ be independent exponential random variables with respective parameters $\{\lambda_r : r \geq 1\}$ no two of which are equal. Find the density function of $S_n = \sum_{r=1}^{n} X_r$. [Hint: Use induction.]

Solution: Let f_n be the density function of S_n. By convolution,

$$f_2(x) = \lambda_1 e^{-\lambda_1 x} \cdot \frac{\lambda_2}{\lambda_2 - \lambda_1} + \lambda_2 e^{-\lambda_2 x} \cdot \frac{\lambda_1}{\lambda_1 - \lambda_2} = \sum_{r=1}^{2} \lambda_r e^{-\lambda_r x} \prod_{s=1, s\neq r}^{2} \frac{\lambda_s}{\lambda_s - \lambda_r}.$$

This leads to the guess that

$$f_n(x) = \sum_{r=1}^{n} \lambda_r e^{-\lambda_r x} \prod_{s=1, s\neq r}^{n} \frac{\lambda_s}{\lambda_s - \lambda_r},\ n \geq 2, \tag{1}$$

which may be proved by induction as follows. Assume that Eq. (1) holds for $n \leq N$. Then,

$$\begin{aligned} f_{N+1}(x) &= \int_0^x \sum_{r=1}^{N} \lambda_r e^{-\lambda_r (x-y)} \lambda_{N+1} e^{-\lambda_{N+1} y} \prod_{s=1, s\neq r}^{N} \frac{\lambda_s}{\lambda_s - \lambda_r} dy \\ &= \sum_{r=1}^{N} \lambda_r e^{-\lambda_r x} \prod_{s=1, s\neq r}^{N+1} \frac{\lambda_s}{\lambda_s - \lambda_r} + A e^{-\lambda_{N+1} x}, \end{aligned}$$

for some constant A. We integrate over x to find that

$$1 = \sum_{r=1}^{N} \prod_{s=1, s\neq r}^{N+1} \frac{\lambda_s}{\lambda_s - \lambda_r} + \frac{A}{\lambda_{N+1}},$$

and Eq. (1) follows with $n = N + 1$ on solving for A.

4. Let X and Y have a bivariate normal density with zero means, variances σ^2, τ^2, and correlation ρ. Show that:

   (a) $\mathbb{E}[X|Y] = \frac{\rho\sigma}{\tau} Y$,

   (b) $\mathrm{var}[X|Y] = \sigma^2(1 - \rho^2)$,

   (c) $\mathbb{E}[X|X + Y = z] = \frac{(\sigma^2 + \rho\sigma\tau)z}{\sigma^2 + 2\rho\sigma\tau + \tau^2}$,

   (d) $\mathrm{var}[X|X + Y = z] = \frac{\sigma^2\tau^2(1-\rho^2)}{\sigma^2 + 2\rho\sigma\tau + \tau^2}$.

Solution: From the representation $X = \sigma\rho U + \sigma\sqrt{1 - \rho^2} V$, $Y = \tau U$, where U and V are independent $N(0, 1)$, we learn that

$$\mathbb{E}[X|Y = y] = \mathbb{E}[\sigma\rho U | U = y/\tau] = \frac{\sigma\rho y}{\tau}.$$

Similarly,

$$\mathbb{E}[X^2|Y = y] = \mathbb{E}[(\sigma\rho U)^2 + \sigma^2(1 - \rho^2)V^2 | U = y/\tau] = \left(\frac{\sigma\rho y}{\tau}\right)^2 + \sigma^2(1 - \rho^2),$$

whence $\mathrm{var}(X|Y) = \sigma^2(1 - \rho^2)$. For parts (c) and (d), simply calculate that $\mathrm{cov}(X, X + Y) = \sigma^2 + \rho\sigma\tau$, $\mathrm{var}(X + Y) = \sigma^2 + 2\rho\sigma\tau + \tau^2$, and

$$1 - \rho(X, X + Y)^2 = \frac{\tau^2(1 - \rho^2)}{\sigma^2 + 2\rho\sigma\tau + \tau^2}.$$